

Digital Electronics Lab Report

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Experiment 4: IC-7483 4-bit adder and IC-74157 quad 2-to-1 Multiplexer

Objectives :

1. To verify the truth table of the 4-bit adder IC-7483
2. To display each of the 4-bit inputs and the output, one at a time by a 7-segment LED display, using quad 2-to-1 Multiplexer IC-74157.

Apparatus :

- IC 7483 — 4-bit binary adder (1)
- IC 74157 — Quad 2-to-1 multiplexer (2)
- IC 7447 — BCD to 7-segment decoder (1)
- 7-segment LED display — Common anode type (1)

- Resistors — 470 Ω or 390 Ω (8)
- Resistors — 2 k Ω (10)
- Breadboard
- DIP switches — 4 or more switches (3 sets)
- Connecting wires
- DC power supply — 5 V (1)

Theory :

This experiment demonstrates the operation of a 4-bit binary adder (IC 7483) and a 2-to-1 multiplexer (IC 74157) along with their use in driving a display unit.

The IC 7483 performs parallel addition of two 4-bit binary numbers and an input carry. It consists of four cascaded full-adder stages that generate sum outputs and a carry output according to binary addition rules. The carry produced in each stage propagates to the next stage, forming a ripple carry structure. Such adders are fundamental components of arithmetic logic units in digital processors.

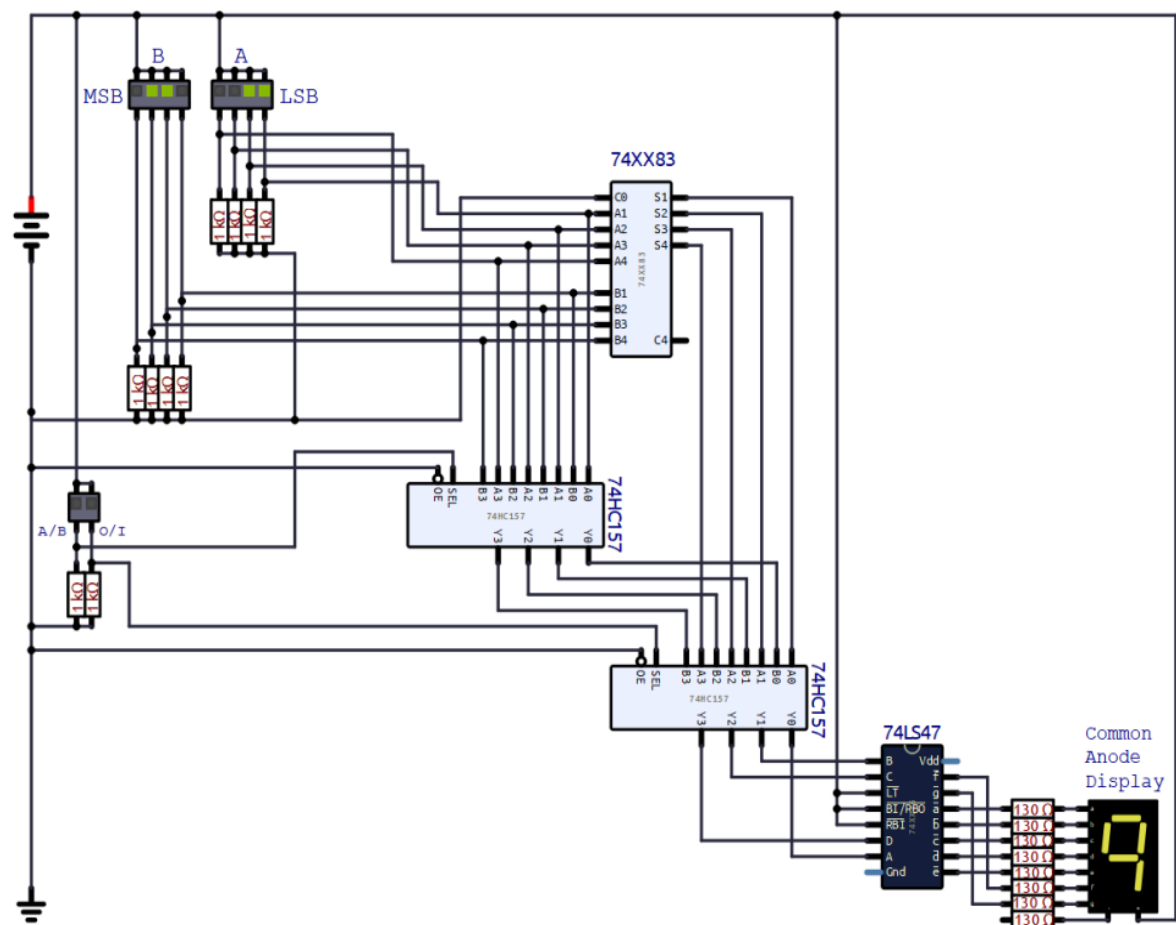
The IC 74157 is a data selector that routes one of two sets of four input signals to a common output based on a select control input. Each channel follows the relation

$$Y = \bar{S}A + SB$$

allowing flexible control of signal flow within digital circuits. Multiplexers are widely used for data routing, resource sharing, and bus selection.

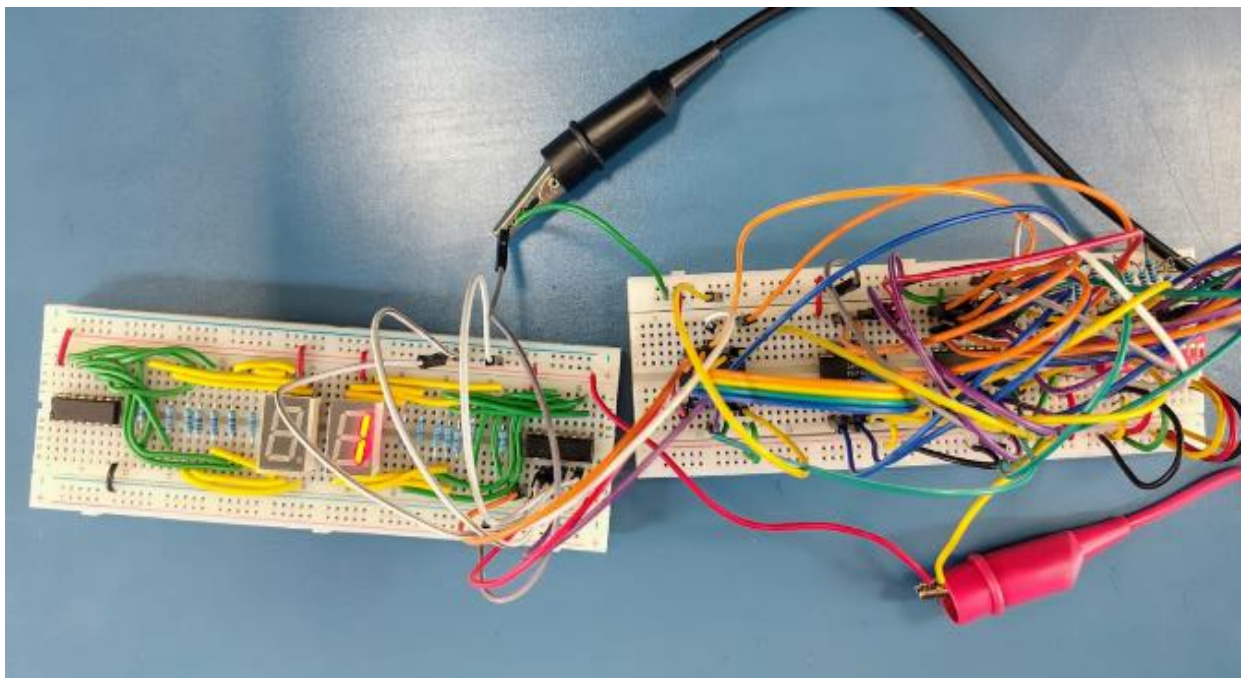
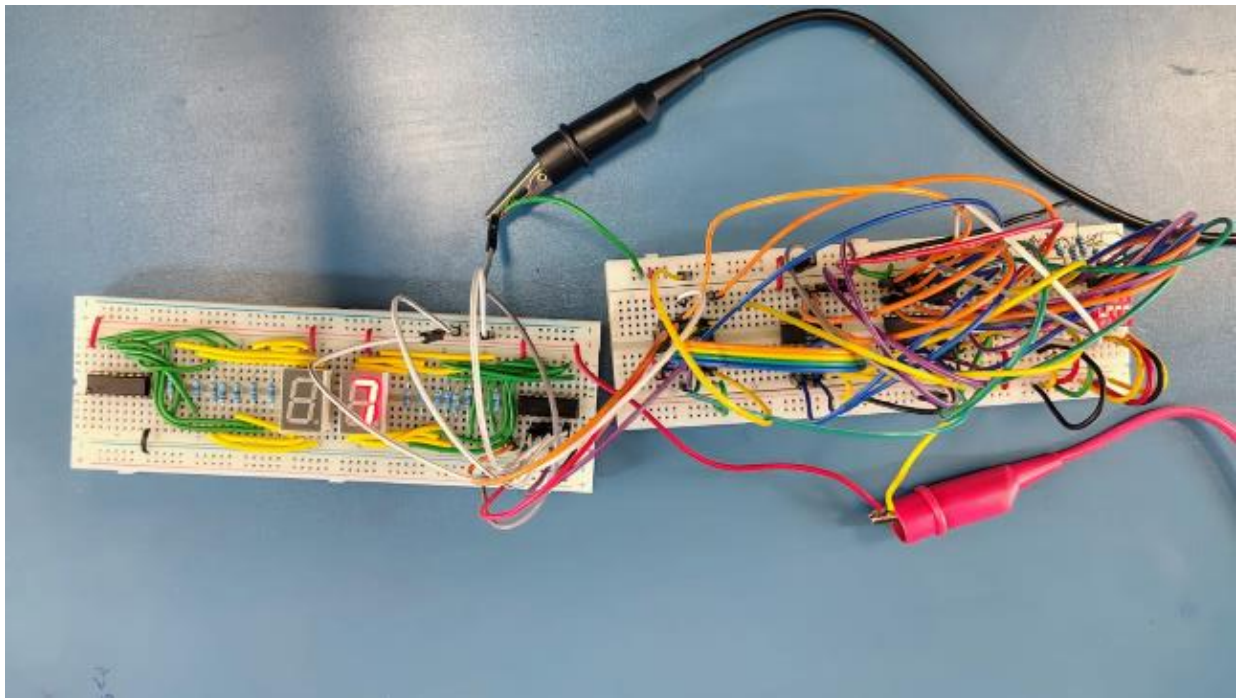
In this setup, the multiplexer selects between different binary values, including operands and adder outputs, and feeds the selected data to a decoder-driven 7-segment display for visualization.

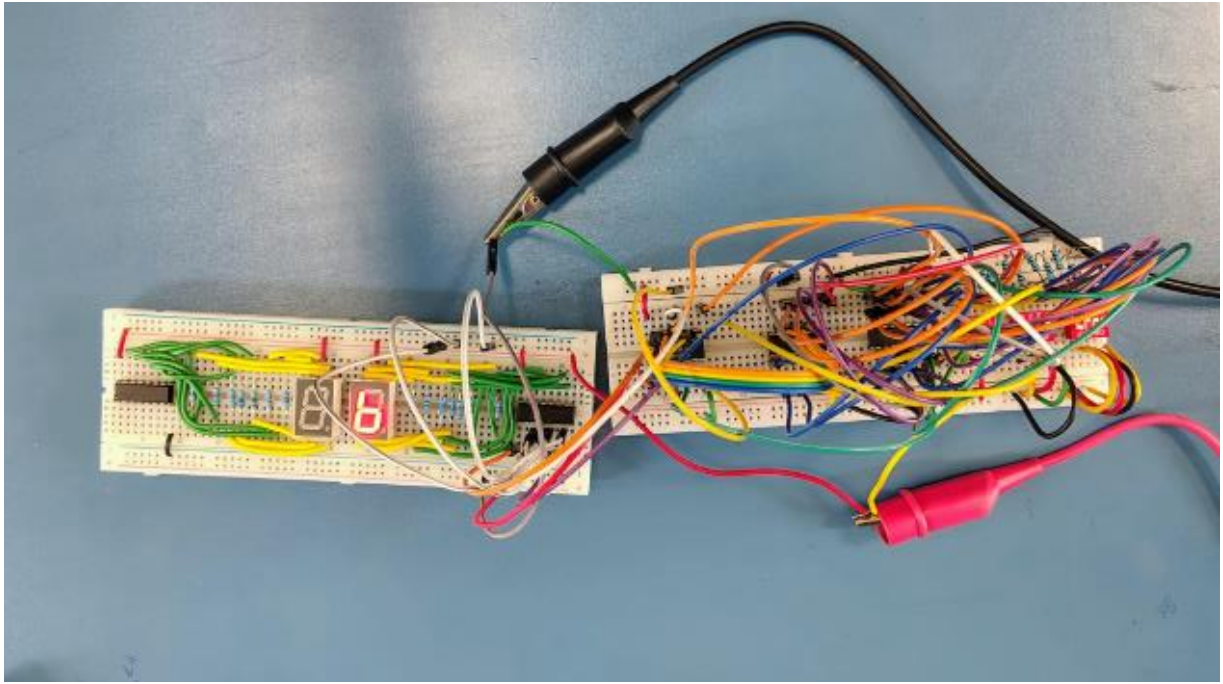
Circuit Diagram :



Output/Observation :

A=6, B=1, Sum=7





Conclusion :

In this experiment, the operation of a 4-bit binary adder (IC 7483) and a 2-to-1 multiplexer (IC 74157) was successfully verified. The adder correctly performed binary addition of two 4-bit inputs along with carry, demonstrating the working of ripple carry adder used in digital computation. The multiplexer effectively selected between different input data sets based on the control signal. By interfacing the outputs with a BCD-to-7-segment decoder and display, the results were visualized, confirming correct functionality.